Scientific Publication in SE, Seminar 1:

Science, Engineering, Community, Sciento- and Bibliometrics

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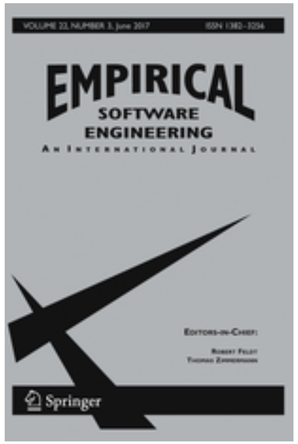
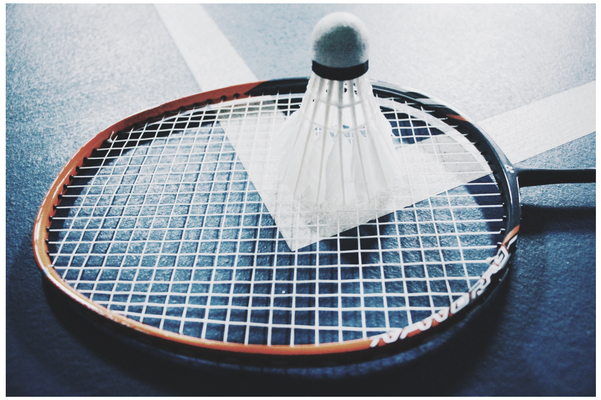
About me

CHALMERS
UNIVERSITY OF TECHNOLOGY

Mittuniversitetet
MID SWEDEN UNIVERSITY





The course is focused on seminars and discussion to help participants form their own views and answers, both on a theoretical and a practical level, to key questions like:

- What is a good scientific study in software engineering (SE)?
- What is a good SE paper?
- In what sense should my evidence support my claims?
- What is the difference between publishing in a journal and at a conference?
- Do different publication venues have different requirements and status? Where should I target my papers?
- How do review processes work, and how do I review papers?
- How is research and how are researchers assessed, and how is that changing?
- What is this thing called science? Is software engineering a science?
- How do AI tools change how we do, write up and review research, and what should we (not) use them for?

The course consists of discussion-based seminars to which participants do some light preparation. The idea is to give hands-on practical advice and show examples of how the software engineering research community functions. Participants bring one of their own papers (finished or in writing), which is used throughout the course, including for analysis with an AI-based claim–evidence tool. Since the course is an initial course, limited in size and scope, we go broad rather than in depth. But, at least, participants get a grounding in key activities and practices and can then go deeper as they see fit.

Course goals

After completion of the course the participants are expected to be able to:

- Read SE research papers and describe their motivation, the methods used and the results achieved
- State the strong and weak points of an SE study, the threats to validity as well as how the study can be improved
- Identify the research paradigm and method used in an SE study and compare it to alternative methods
- Write good and constructive reviews of scientific papers in software engineering
- Apply review templates objectively even on their own research studies and papers
- Check whether the claims of a paper, including their own, are supported by its evidence
- Select relevant publication targets for their research
- Describe the strongest publication targets for SE in general as well as for their sub-area
- Write and update publication strategies for their research
- Define bibliometrics used to rank publication targets, journals and researchers, and discuss their limitations as well as current efforts to reform research assessment
- Discuss the opportunities and risks of using AI tools in research, writing and reviewing, and state their own policy for such use

	Monday 13:15–17:00	Tuesday 09:15–16:00	Wednesday 09:15–15:00
Theme	Science and the publication system	Judging papers, others' and your own	Trustworthy evidence and the future of research
Morning		How editors and PCs decide · Reading and reviewing papers	Research methods and validity · AI, evidence and trust
Lunch		12:00–13:00	12:00–13:00
Afternoon	Science vs. engineering, evidence · Metrics and reform of research assessment · The publication system under strain	Do my claims follow from my evidence? (AI claim–evidence analysis) · Paper clinics	AI and the future of science · Your AI-use policy · Your publication strategy, next steps
Evening	Course dinner		
Preparation	Day 1 prep	Day 2 prep	Day 3 prep

Ground Rules / Format

- We are colleagues having a discussion to broaden our views
- Everyone is active in discussions, everyone counts
- There is seldom a single (correct) answer
- No question is dumb; we learn by questioning, reflecting & sharing
- Format:
  - Initial discussion on prep questions
  - Robert shares some slides and we discuss the questions further as we go along
  - Polish is less important than content & repetition often good; slides are gonna be messy :)

Disclaimer

- I'll give examples based on my experience and my views
  - might not be the views of all faculty / your supervisor
  - clearly not the views of Springer, EMSE, ESEM SC or other organisations I'm part of or work with
- I'm sorry if I sometimes come across as cynical
  - sometimes hard to stay "idealistic" after a work life in academia

A key divide: Ideal vs Pragmatic View

Acknowledgements

While this is based on courses I have given at Chalmers and BTH before I got most of it from others and colleagues over the years.

In particular I want to thank Daniel Mendez (prof at BTH/Sweden & at Fortiss in Munich/Germany) for his slides on science, Empirical SE & methods

Ok to apply LLM-based tool on your pdfs?

So far only run on the already published papers some of you selected.

For better AI-based paper analysis we cannot use current open weight LLMs, so do you allow use of commercial LLMs?

What is science? What is engineering? Importance of community?

Let's discuss!

- What is science?

- What is engineering?

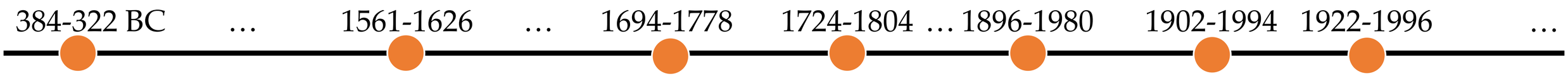
- How do they differ?

- How does a research study differ from an improvement project in a company?

“Science” wasn’t built in a day…

> Science is understood as the human undertaking for the search of knowledge (through systematic application of scientific methods)

→ Needs to be considered in a historical context
→ Increased understanding of scientific practice (and what science eventually is)



Aristoteles

- Search for laws and reasoning for phenomena
- Understanding the nature of phenomena

Bacon

- Progress of knowledge of nature (reality)
- Draw benefits from knowledge growth

Voltaire

- Emancipation from gods and beliefs

Kant

- System of Epistemology (theory of knowledge)

Piaget

- Era of constructivism

Popper

- Falsification as demarcation criterion
- Birth of null hypothesis testing

Kuhn

- Paradigm shifts
- Scientific Revolutions

“Science” wasn’t built in a day…

1922-1974 — **Lakatos**

- Competing Research Programs
- Progressive RP = predicts novel facts
- Degenerative RP = protects it’s core assumptions

1924-1994 — **Feyerabend**

- Against (single) Method
- Epistemological Anarchism (Pluralism & Creativity)

1941- — **Van Fraassen**

- Constructive Empiricism = Empirical Adequacy
- Focus on models
- Theories = family of models

1944- — **Cartwright**

- Pragmatic focus
- Causation as basis

How the laws of physics lie
N Cartwright, E McMullin
American Journal of Physics 52 (5), 474-476

1944- — **Longino**

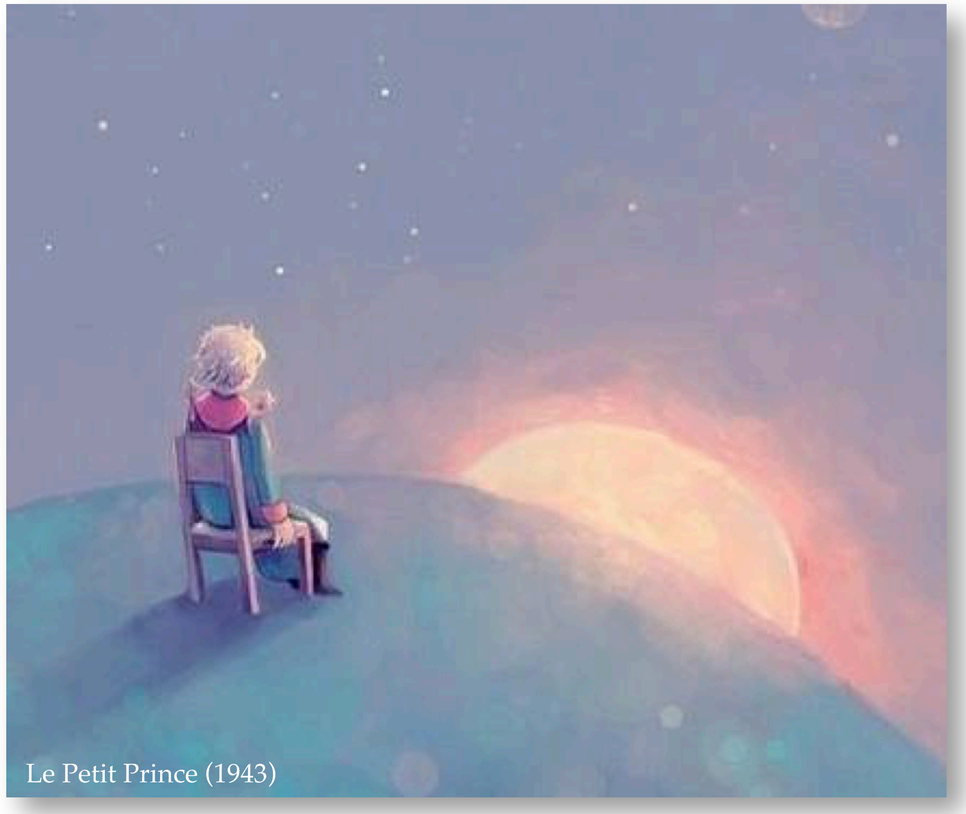
- Social Nature of Science/Knowledge
- Inclusive Objectivity

…

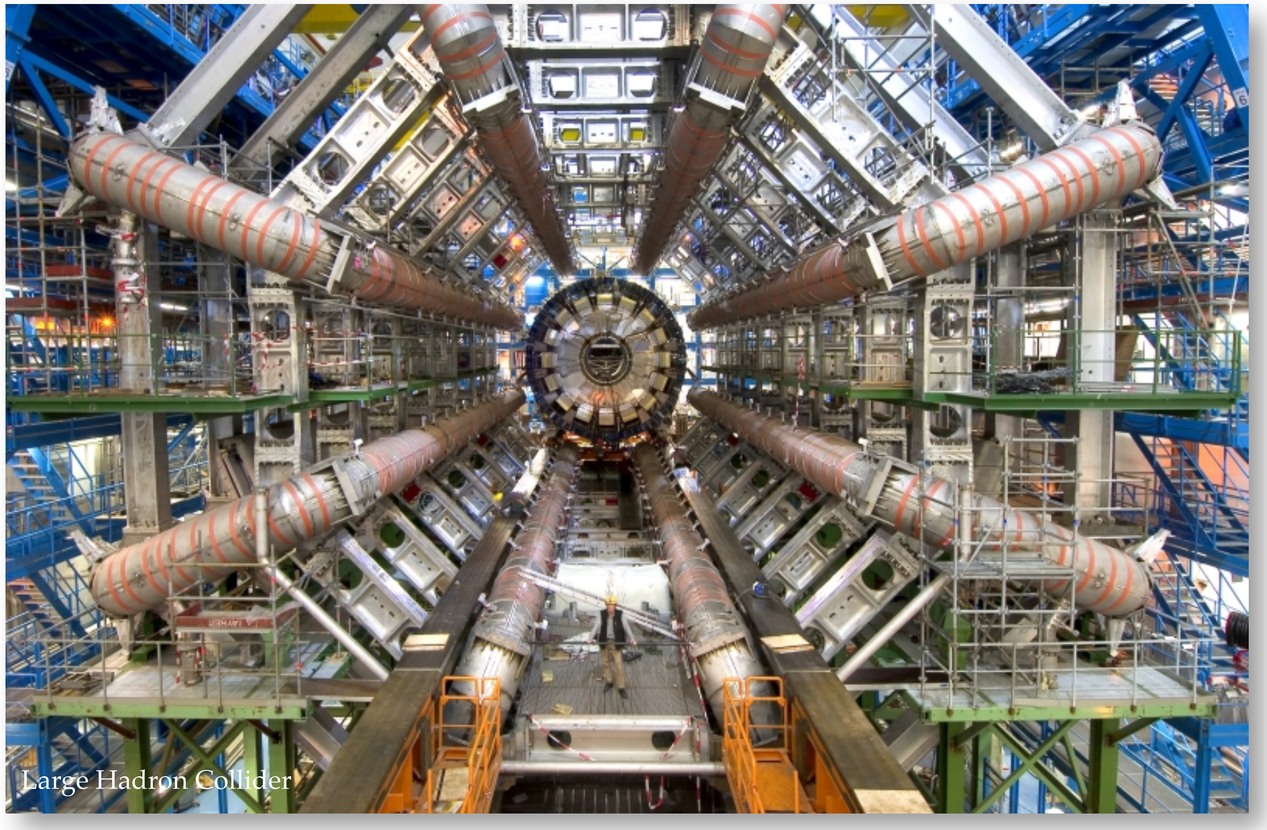
Scientific practices and research methods have changed over time, the role of empiricism* not

384-322 BC

Today



Le Petit Prince (1943)



Large Hadron Collider

* Gaining knowledge through sensory experiences

Scientific knowledge and practice

> Scientific knowledge is the portrait of
> our understanding of reality (via scientific theories).

Necessary postulates for scientific practice (selected):

- There are certain rules, principles, and norms for scientific practices
  - Rationalism: Reasoning by argument / logical inference / mathematical proof
  - Empiricism: Reasoning by sensory experiences (case studies, experiments,…)
- There is nothing absolute about truth
- There is a scientific community to judge about the quality of empirical studies

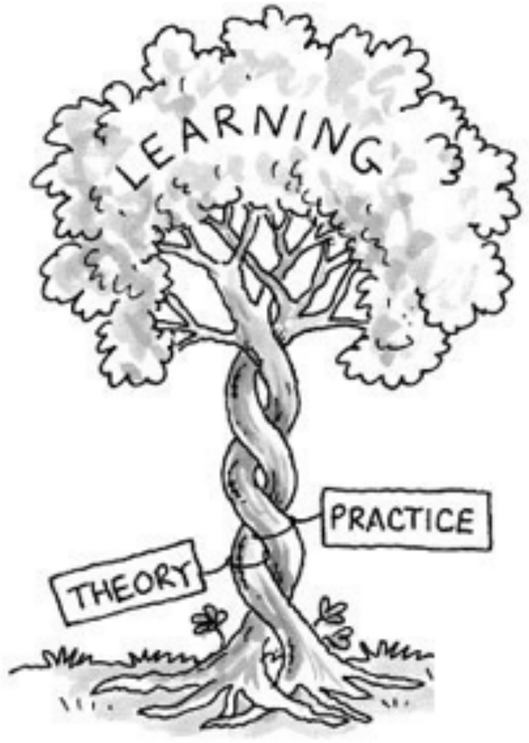
One view:

Empirical Software Engineering

> The ultimate goal of Empirical Software Engineering is to advance our body of knowledge by building and evaluating theories.

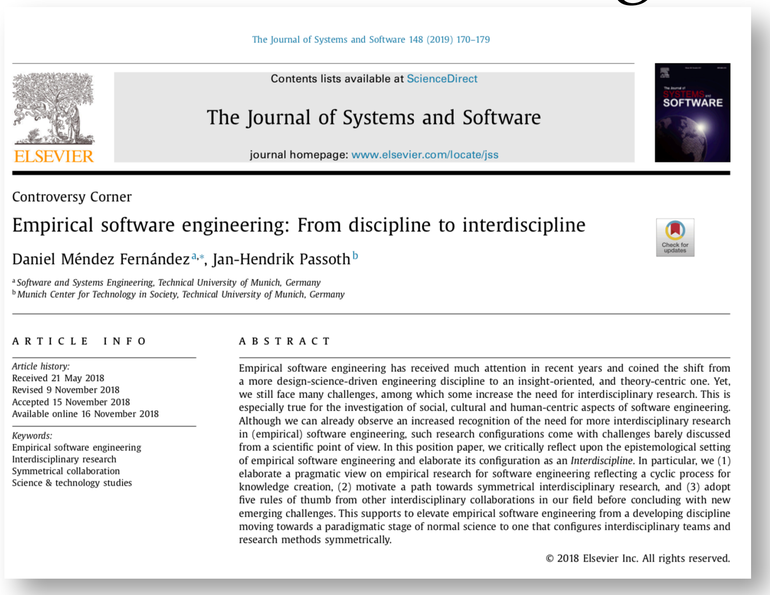
Relevance from a theoretical and practical perspective:

- Reason about the discipline and (e.g. social) phenomena involved
- Recognise and understand limitations and effects of artefacts (e.g. by evaluating technologies, techniques, processes, models, etc.) in their practical contexts



From real world phenomena to theories and back: The empirical life cycle

Further reading and outlook



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Controversy Corner

Empirical software engineering: From discipline to interdiscipline

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ABSTRACT

Empirical software engineering has received much attention in recent years and coined the shift from a more design-science-driven engineering discipline to an insight-oriented, and theory-centric one. Yet, we still face many challenges, among which some increase the need for interdisciplinary research. This is especially true for the investigation of social, cultural and human-centric aspects of software engineering. Although we can already observe an increased recognition of the need for more interdisciplinary research in (empirical) software engineering, such research configurations come with challenges barely discussed from a scientific point of view. In this position paper, we critically reflect upon the epistemological setting of empirical software engineering and elaborate its configuration as an *Interdiscipline*. In particular, we (1) elaborate a pragmatic view on empirical research for software engineering reflecting a cyclic process for knowledge creation, (2) motivate a path towards symmetrical interdisciplinary research, and (3) adopt five rules of thumb from other interdisciplinary collaborations in our field before concluding with new emerging challenges. This supports to elevate empirical software engineering from a developing discipline moving towards a paradigmatic stage of normal science to one that configures interdisciplinary teams and research methods symmetrically.

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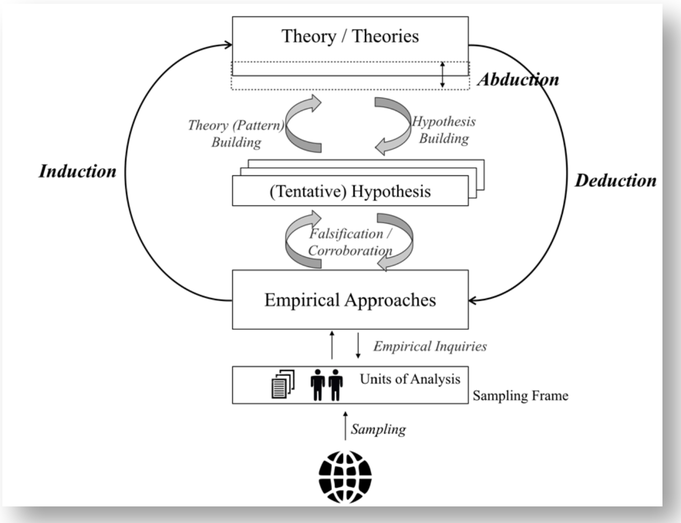


Preprint: https://arxiv.org/abs/1805.08302

- Epistemological setting of Empirical Software Engineering
- Theory building and evaluation
- Challenges in Empirical Software Engineering

duction

(Creative) Synthesis of an explanatory case from a general rule and a particular result (observation)

Deduction

Application of a general rule to a particular case, inferring a specific result

pling Frame

Source: Méndez and Passoth: Empirical Software Engineering: from Discipline to Interdiscipline, 2018.

What you believe and why

Let's discuss!

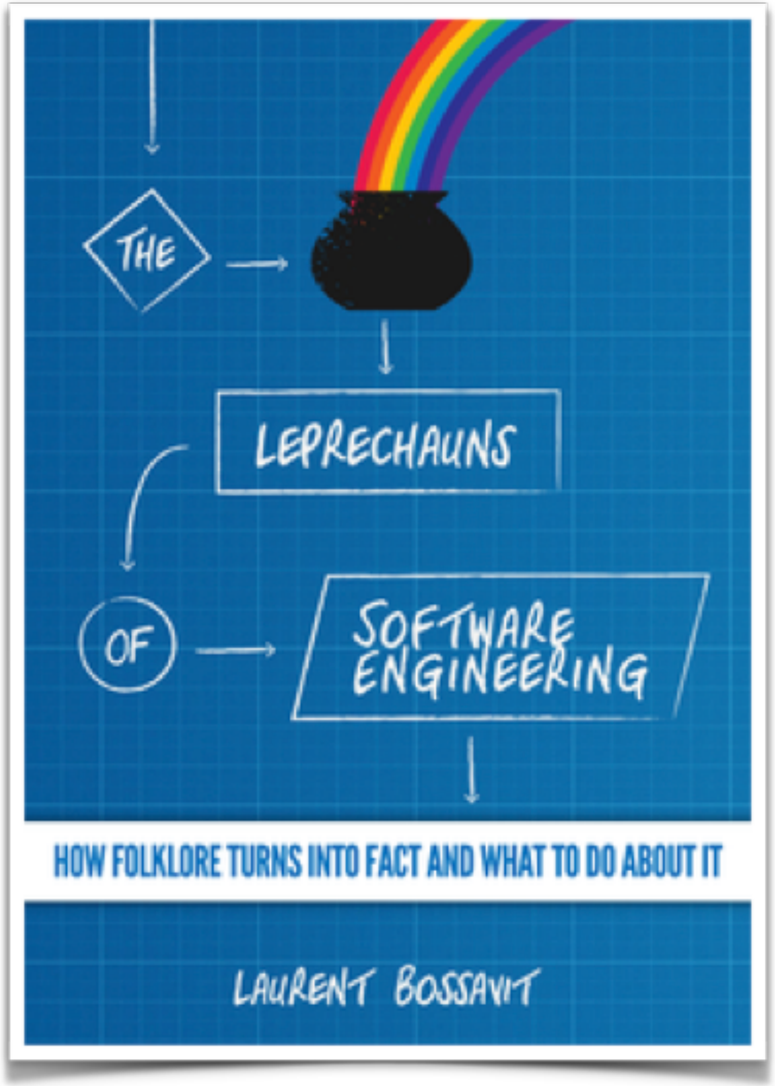
- Name one belief that is common in SE (in practice or in your research area)

- What evidence do you actually know of that supports it?

- Would that evidence convince a sceptic?

Conventional Wisdom in SE

- Emerge from times where claims by authorities were treated as “facts”
- Reasons manifold:
  - Lack of empirical awareness
  - Neglecting particularities of practical contexts
  - Neglecting relation to existing evidence
  - No proper citations (one side of the medal or over-conclusions)
  - Lack of data
  - …



Exemplary conventional wisdom:
Go To statements considered harmful

1968

Edgar Dijkstra: Go T

Go To Statement Consider

Key Words and Phrases: go to
branch instruction, conditional
itive clause, program intelligibil
CR Categories: 4.22, 5.23, 5.24

EDITOR:

For a number of years I have be
that the quality of programmers
density of **go to** statements in the
recently I discovered why the use
disastrous effects, and I became
ment should be abolished from al
languages (i.e. everything except,
At that time I did not attach to
covery; I now submit my consid
in very recent discussions in whic
been urged to do so.

My first remark is that, altho
ends when he has constructed a
taking place under control of hi
matter of his activity, for it is this
the desired effect; it is this proces
has to satisfy the desired specificat
been made, the "making" of the
gated to the machine.

2015

An Empirical Study of Goto in C Code from GitHub Repositories

Meiyappan Nagappan[1], Romain Robbes[2], Yasutaka Kamei[3], Éric Tanter[2], Shane McIntosh[4], Audris Mockus[5], Ahmed E. Hassan[6]
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ABSTRACT

It is nearly 50 years since Dijkstra argued that `goto` obscures the flow of control in program execution and urged programmers to abandon the `goto` statement. While past research has shown that `goto` is still in use, little is known about whether `goto` is used in the unrestricted manner that Dijkstra feared, and if it is 'harmful' enough to be a part of a post-release bug. We, therefore, conduct a two part empirical study - (1) qualitatively analyze a statistically representative sample of 384 files from a population of almost 250K C programming language files collected from over 11K GitHub repositories and find that developers use `goto` in C files for error handling (80.21±5%) and cleaning up resources at the end of a procedure (40.36 ± 5%); and (2) quantitatively analyze the commit history from the release branches of six OSS projects and find that no `goto` statement was removed/modified in the post-release phase of four of the six projects. We conclude that developers limit themselves to using `goto` appropriately in most cases, and not in an unrestricted manner like Dijkstra feared, thus suggesting that `goto` does not appear to be harmful in practice.

Harmful [11]. This is one of the many works of Dijkstra that is frequently discussed by software practitioners [25] and researchers alike (more than 1,300 citations according to Google Scholar and almost 4000 citations according to ACM Digital Library as of Aug 15, 2014). This article has also resulted in a slew of other articles of the type *global variables considered harmful* [32], *polymorphism considered harmful* [24], *fragmentation considered harmful* [16], among many others. In fact, Meyer claims that as of 2002, there are thousands of such articles, though most are not peer-reviewed [15].

Indeed, Dijkstra's article [11] has had a tremendous impact. Anecdotally, several introductory programming courses instruct students to avoid `goto` statements solely based on Dijkstra's advice. Marshall and Webber [19] warn that when programming constructs like `goto` are forbidden for long enough, they become difficult to recall when required.

Dijkstra's article on the use of `goto` is based on his desire to make programs verifiable. The article is not just an opinion piece; as Koenig points out [7], Dijkstra provides strong *logical* evidence for why `goto` statements can introduce problems in software.

1969

O Considered Harmful"
lered Harmful

ost-noted item ever pub-
in *Communications* was a
from Edsger W. Dijkstra
d "Go To Statement Con-
d Harmful" [1] which at-
ed to give a reason why the
statement might be harm-
though the argument was
nic and unconvincing, its
ems to have become fixed
mind of every programming
er and methodologist. Con-
tly, the notion that the
is harmful is accepted al-
niversally, without question
bt. To many people, "struc-
programming" and "GOTO-
ogramming" have become
synonymous.

1987

"'GOTO Considered Harmful' Considered Harmful" Considered Harmful?

I enjoyed Frank Rubin's letter ("'**GOTO** Considered Harmful' Considered Harmful," March 1987, pp. 195–196), and welcome it as an opportunity to get a discussion started. As a software engineer, I have found it interesting over the last 10 years to write programs both with and without **GOTO** statements at key points. There are cases where adding a **GOTO** as a quick exit from a deeply nested structure is convenient, and there are cases where revising to eliminate the **GOTO** actually simplifies the program.

"We conclude that developers limit themselves to using goto appropriately, [not] like Dijkstra feared, [thus] goto does not appear to be harmful in practice."

- Public exchange based on reasoning by argument (rationalist arguments)...
- … finally challenged by one single empirical study.

[1] Edsger Dijkstra . Go To Statement Considered Harmful. Communications of the ACM, 1968.
[2] Frank Rubin. "GOTO Considered Harmful" Considered Harmful. Communications of the ACM, 1969.
[3] Donald Moore et al. " 'GOTO Considered Harmful' Considered Harmful" Considered Harmful?" Communications of the ACM, 1987.
[4] Nagappan et al. An empirical study of goto in C code from GitHub repositories, 2015.

Wohlin's Types of Evidence Levels

Category	Typical examples
Strong evidence	Independent, peer-reviewed experiments with industrial participants or cross-company multiple-case studies
Evidence	Independent experiments with less representative participants or several case studies within a company
Circumstantial evidence	Single case studies, cross-company surveys or experiments involving vested interests
Third-party claim	Independent experience reports, lessons learned or single-company surveys
First- or second-party claim	Claims from inventors or others with vested interests

Us if we are evaluating our own tools!?

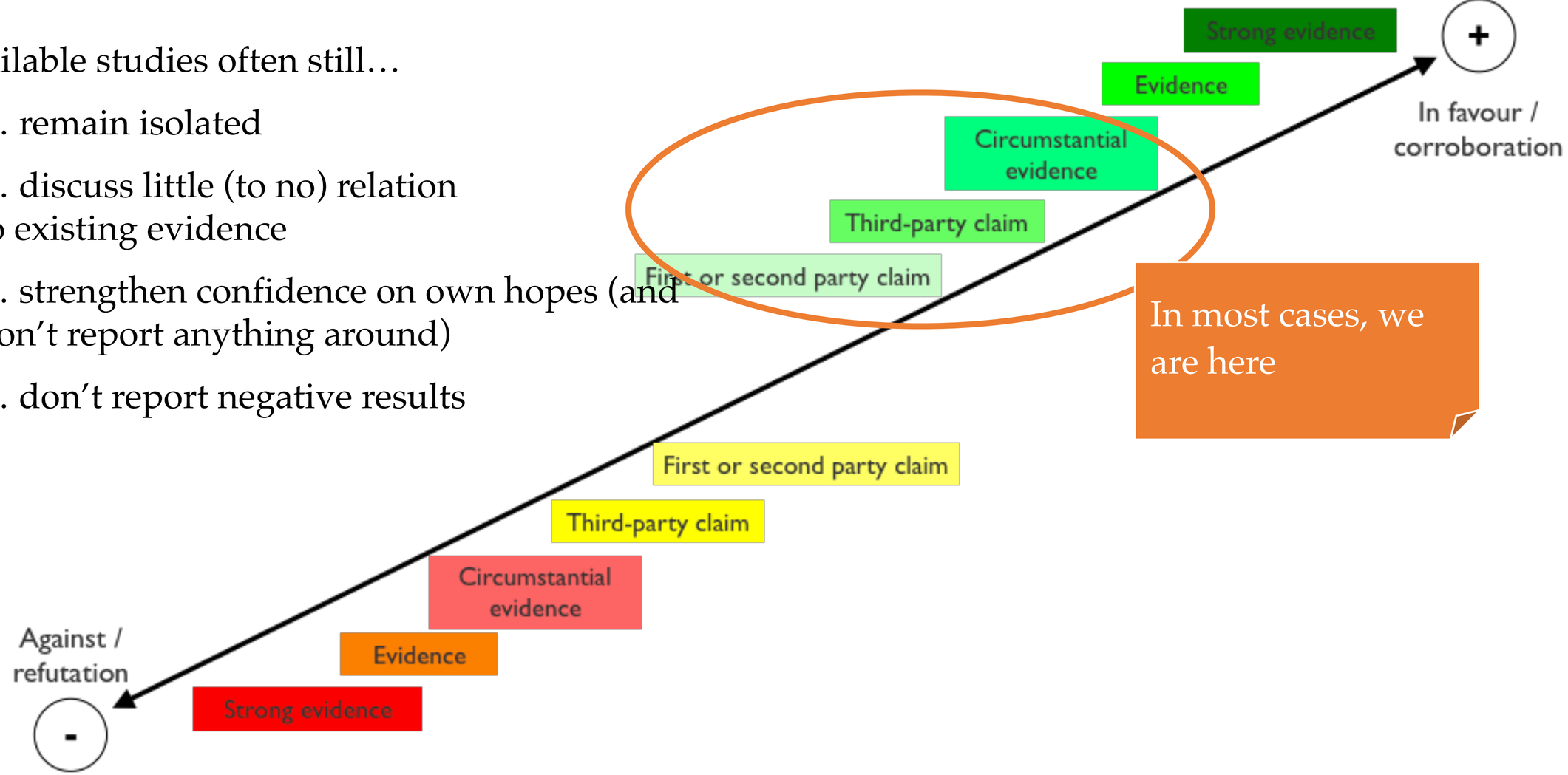
- Since evidence can be positive or negative (supporting or opposing), and adding "inconclusive" one can map evidence to an 11-point scale

Source (levels of evidence): Wohlin. An Evidence Profile for Software Engineering Research and Practice, 2013.

Current state of evidence in Software Engineering

Available studies often still…

- … remain isolated
- … discuss little (to no) relation to existing evidence
- … strengthen confidence on own hopes (and don't report anything around)
- … don't report negative results



Source (levels of evidence): Wohlin. An Evidence Profile for Software Engineering Research and Practice, 2013.

Learn evidence assessment from other fields? (Classic tactic: Check Medicine)

Strength of Evidence in Systematic Reviews in Software Engineering

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Already 2008!

Table 4. Strength of evidence i

High	Further research is confidence in the e
Moderate	Further research is impact on our conf and may change th
Low	Further research is important impact o estimate of effect a estimate.
Very low	Any estimate of eff

Table 3. Quality assessment of 33 SE studies [15]

1. Is the paper based on research (or is it merely a "lessons learned" report based on expert opinion)? (33)
2. Is there a clear statement of the aims of the research? (31)
3. Is there an adequate description of the context in which the research was carried out? (33)
4. Was the research design appropriate to address the aims of the research? (30)
5. Was the recruitment strategy appropriate to the aims of the research? (8)
6. Was there a control group with which to compare treatments? (10)
7. Was the data collected in a way that addressed the research issue? (26)
8. Was the data analysis sufficiently rigorous? (25)
9. Has the relationship between researcher and participants been adequately considered? (1)
10. Is there a clear statement of findings? (33)
11. Is the study of value for research or practice? (33)

y decrease or increase the strength of evidence [20]

ase the strength of evidence:

rious (–2) limitations to study quality
y (–1)
2) uncertainty about directness
ta (–1)
rting bias (–1)

se the strength of evidence:

ociation (RR > 2 or RR < 0.5) based on
m two or more observational studies,
unders (+1)
f association (RR > 5 or RR < 0.2)
ce with no major threats to validity (+2)
sponse gradient (+1)
rs would have reduced the effect (+1)

Learn evidence assessment from other fields? (Classic tactic: Check Medicine)

Strength of Evidence in Systematic Reviews in Software Engineering

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- In the agile review:
  - only 8 of 33 studies had appropriate recruitment strategy, and
  - only 10 had a comparison group, and
  - just 1 considered researcher-participant relationship (reflexivity, researcher bias
- Also notes (from Sjöberg et al) that in 15 differentiated replications:
  - 7 of 8 replications by original authors confirmed original results, but
  - only 1 of 7 by independent authors confirmed original results.
- GRADE in SE:
  - Mostly observational studies => starts out low evidence
  - Serious quality limitations, inconsistent results and major uncertainty about directness (students instead of industry practitioners)
  - Conclusion: Overal strength of evidence is **VERY LOW**

GRADE - grading evidence in medicine

Check for updates

Why Core GRADE is needed: introduction to a new series in *The BMJ*

Gordon Guyatt,[1,2,3] Monica Hultcrantz,[4,5] Thomas Agoritsas,[1,3,6] Alfonso Iorio,[1,2] Per Olav Vandvik,[3,7] Victor M Montori[8,9]

#	Title	DOI
Intro	Why Core GRADE is needed: introduction to a new series in The BMJ	10.1136/bmj-2024-081902
1	Overview of the Core GRADE approach	10.1136/bmj-2024-081903
2	Choosing the target of certainty rating and assessing imprecision	10.1136/bmj-2024-081904
3	Rating certainty of evidence: assessing inconsistency	10.1136/bmj-2024-081905
4	Rating certainty of evidence: risk of bias, publication bias, and reasons for rating up certainty	10.1136/bmj-2024-083864
5	Rating certainty of evidence: assessing indirectness	10.1136/bmj-2024-083865
6	Presenting the evidence in summary of findings tables	10.1136/bmj-2024-083866
7	Principles for moving from evidence to recommendations and decisions	10.1136/bmj-2024-083867

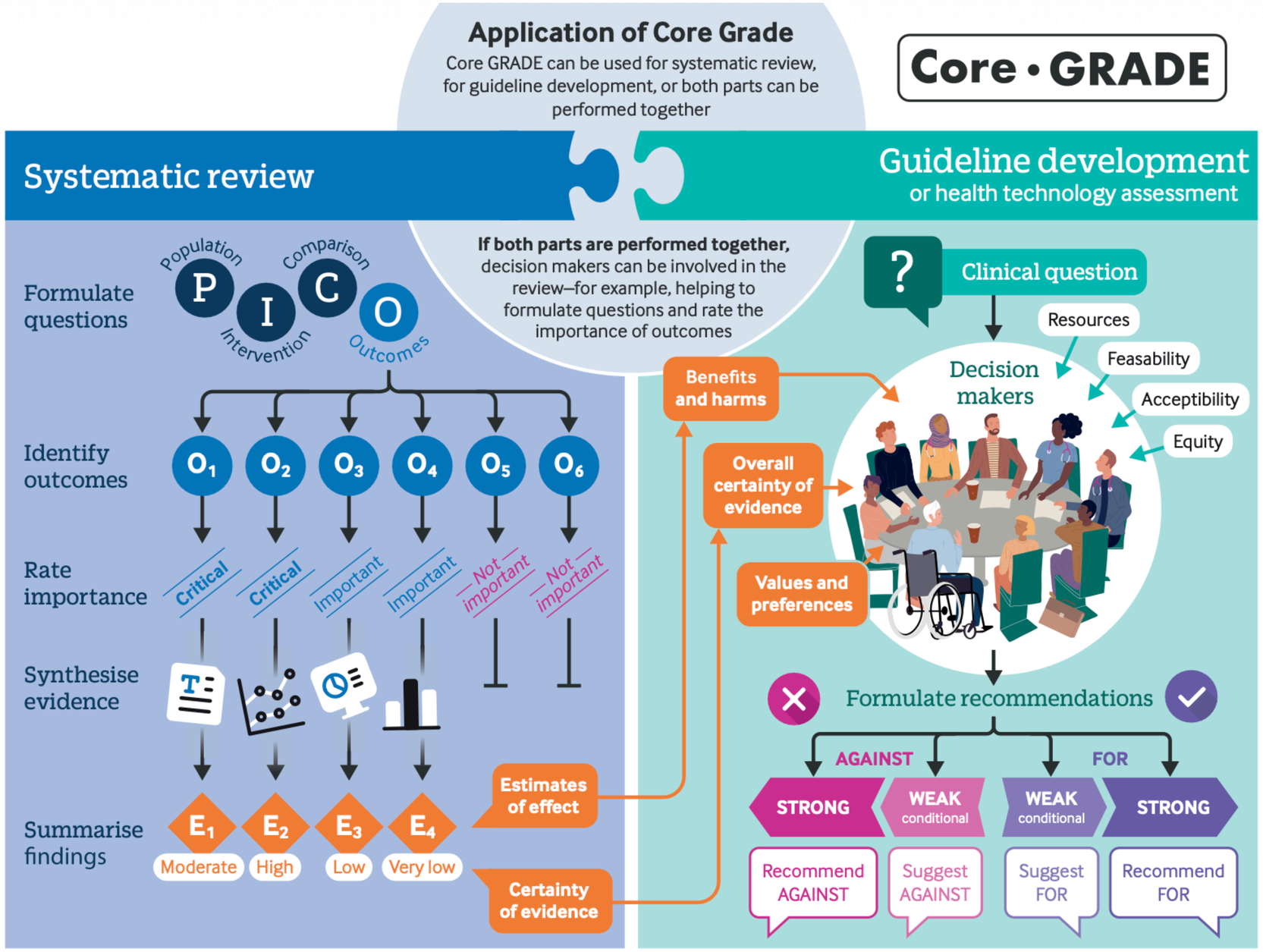


Fig 1 | Schematic overview of Core GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach

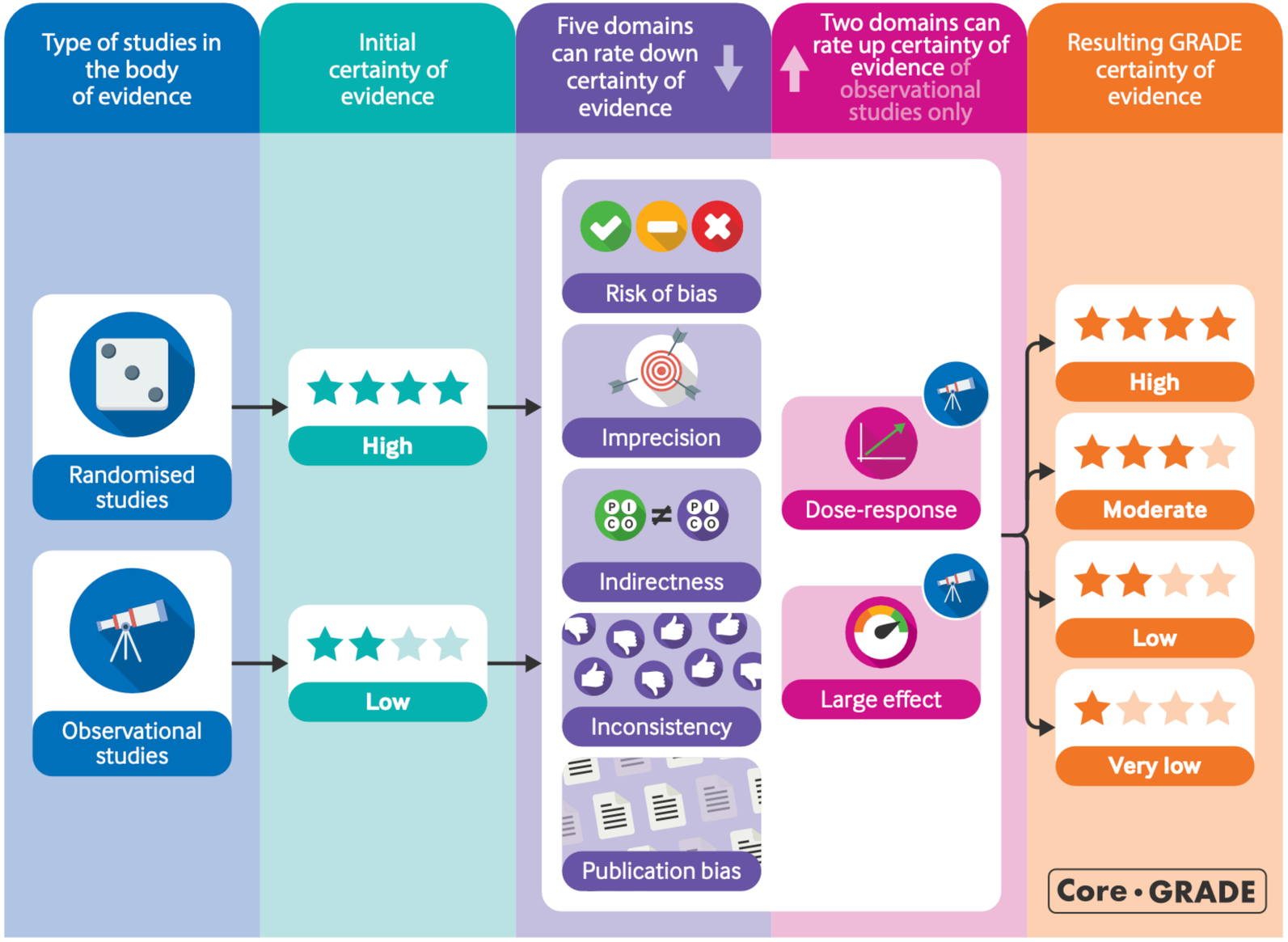


Fig 2 | Core GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach to rating certainty of evidence in intervention effects

GRADE: Still subjective!

Limitations of the Core GRADE approach

We see only one serious limitation in the Core GRADE approach, and that is the subjectivity involved in virtually every important decision in rating certainty of evidence and moving from evidence to recommendations. This need for subjective judgments would, however, be inevitable for any useful system considering certainty of evidence or providing an approach to move from evidence to recommendations. The alternative—a mechanical approach offering never-to-be-broken rules—would result in many poor judgments and prove unfeasible and unacceptable.

But is GRADE applicable to SE?
Qualitative studies etc?

Medicine to the rescue again: GRADE-CERQual!

INTRODUCTION Open Access

Applying GRADE-CERQual to qualitative evidence synthesis findings: introduction to the series

Simon Lewin[1,2*], Andrew Booth[3], Claire Glenton[1], Heather Munthe-Kaas[1], Arash Rashidian[4,5], Megan Wainwright[6], Meghan A. Bohren[7], Özge Tunçalp[7], Christopher J. Colvin[6], Ruth Garside[8], Benedicte Carlsen[9], Etienne V. Langlois[10] and Jane Noyes[11]

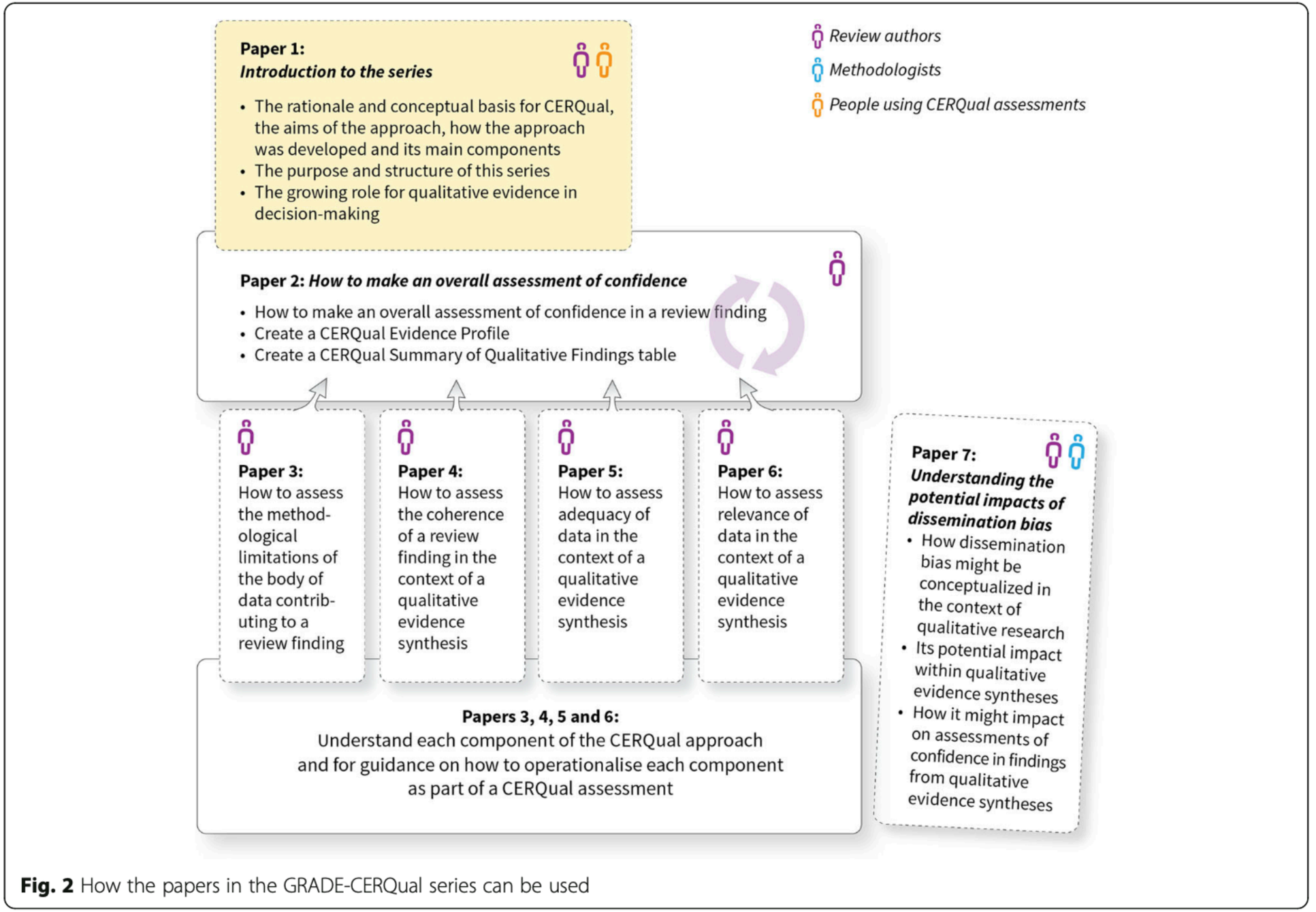


Fig. 2 How the papers in the GRADE-CERQual series can be used

Social objectivity?? Subtle realist position!

Assumptions underlying the development of CERQual

As a pragmatic approach, CERQual makes several assumptions and acknowledgements regarding ongoing methodological debates in the field of qualitative research:

- We acknowledge that some within the qualitative research community have argued that synthesising data across multiple qualitative studies challenges the integrity of the contributing primary studies and that findings from this synthesis process may therefore not be trustworthy (e.g., [19–21]). However, in our approach to qualitative evidence synthesis and the development of CERQual, we have adopted the 'subtle realist' position [22] which maintains that the existence of phenomena does not depend on our subjective perceptions of them. In other words, social reality is not entirely constructed. Based on this, we suggest that synthesis can potentially provide a deeper understanding of a phenomenon than is achievable by any one single study, that this understanding can be viewed as trustworthy, and that it is therefore desirable to synthesise data from multiple qualitative studies. Others have made similar arguments in relation to ethnography, noting that comparisons of different ethnographic studies 'are fruitful because they lead to empirical generalisations, they expose analytical problems, and they allow for falsification of hypotheses' ([23] page 207).

GRADE-CERQual

- **Confidence** = the extent to which a review finding is a reasonable representation of the phenomenon of interest
- **Per finding, not per study!**
  - Collect findings from multiple studies and then
  - Judge confidence in four levels per each of 4 components:
    - **Methodological limitations** = design or conduct of the contributing primary studies
    - **Coherence** = how clear and cogent the fit is between the primary data and the review finding
    - **Adequacy of data** = richness and quantity of the supporting data
    - **Relevance** = how well the supporting evidence applies to the context
  - **Output:** Summary of Qualitative Findings table

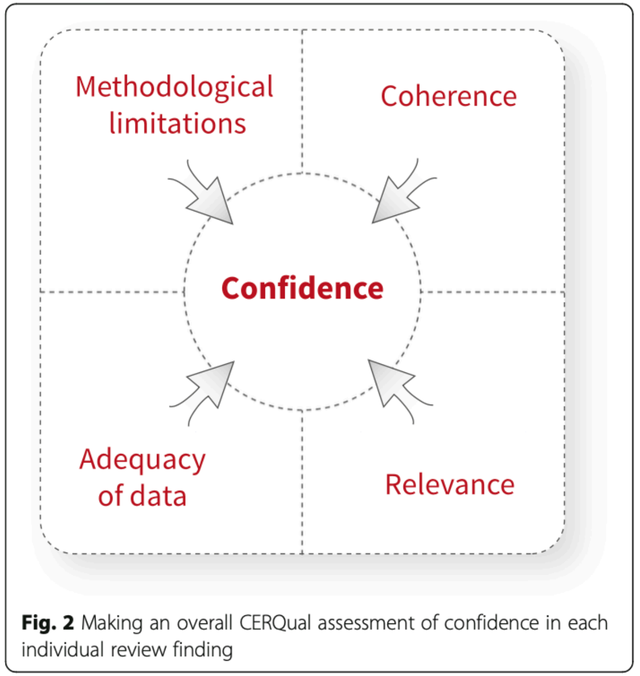


Fig. 2 Making an overall CERQual assessment of confidence in each individual review finding

Discussion continue on how to slice/add components!

Table 3 CERQual Evidence Profile—Example A

Summary of review finding	Studies contributing to the review finding	Methodological limitations	Coherence	Adequacy	Relevance[a]	*CERQual assessment of confidence in the evidence*	Explanation of CERQual assessment
1. Use of force: Women across the world reported experiencing physical force by health providers during childbirth. In some cases, women reported specific acts of violence committed against them during childbirth, but women often referred to these experiences in a general sense and alluded to beatings, aggression, physical abuse, a rough touch and use of extreme force. Pinching, hitting and slapping, either with an open hand or an instrument were the most commonly reported specific acts of physical violence.	6, 9, 10, 13, 21, 61, 67, 68, 73, 75, 77, 80, 84, 86, 87, 91, 96, 97	Moderate methodological limitations (6 studies with minor, 6 studies with moderate (unclear recruitment and sampling), and 3 studies with serious methodological limitations (unclear reflexivity, insufficiently rigorous data analysis))	No or very minor concerns about coherence	No or very minor concerns about adequacy	Minor concerns about relevance (5 studies with direct relevance, 8 studies with partial relevance, and 1 study with unclear relevance. 15 studies total from 10 countries, including 1 high income, 2 middle income and 7 low income countries. Geographical spread: 2 studies in Asia, 1 study in Europe, 1 study in LAC, 1 study in MENA, 1 study in South America, and 8 studies from sub-Saharan Africa.)	High confidence	15 studies with moderate methodological limitations. Data from 10 countries across all geographical regions, but predominantly sub-Saharan Africa. No or very minor concerns about coherence and adequacy.
2. Physical restraint: Women in Tanzania and Brazil reported physical restraint during childbirth through the use of bed restraints and mouth gags.	86, 97	Moderate methodological limitations (1 study with minor and 1 study with serious methodological limitations (inappropriate research design, no reflexivity, unclear ethical considerations))	Minor concerns about coherence (Some concerns about the fit between the data from primary studies and the review finding)	Serious concerns about adequacy (2 studies in total with limited, thin data)	Moderate concerns about relevance (2 studies with partial relevance from 2 countries)	Very low confidence	Two studies (Tanzania and Brazil) with moderate methodological limitations. Limited, thin data from 2 countries. Minor concerns about coherence but limited data available.

Review findings taken from [12] and adapted to fit the context of this article. The review findings presented here are drawn from the wider thematic synthesis undertaken for this review. The themes identified were summarised into evidence statements, as illustrated in this table. The methods are described in more detail in [12]

[a]When describing relevance judgements, consider the following prompts to help elucidate 'partial' and/or 'unclear' relevance: phenomenon of interest, population (including subgroups), setting, place, intervention, findings

Did SE take on GRADE??

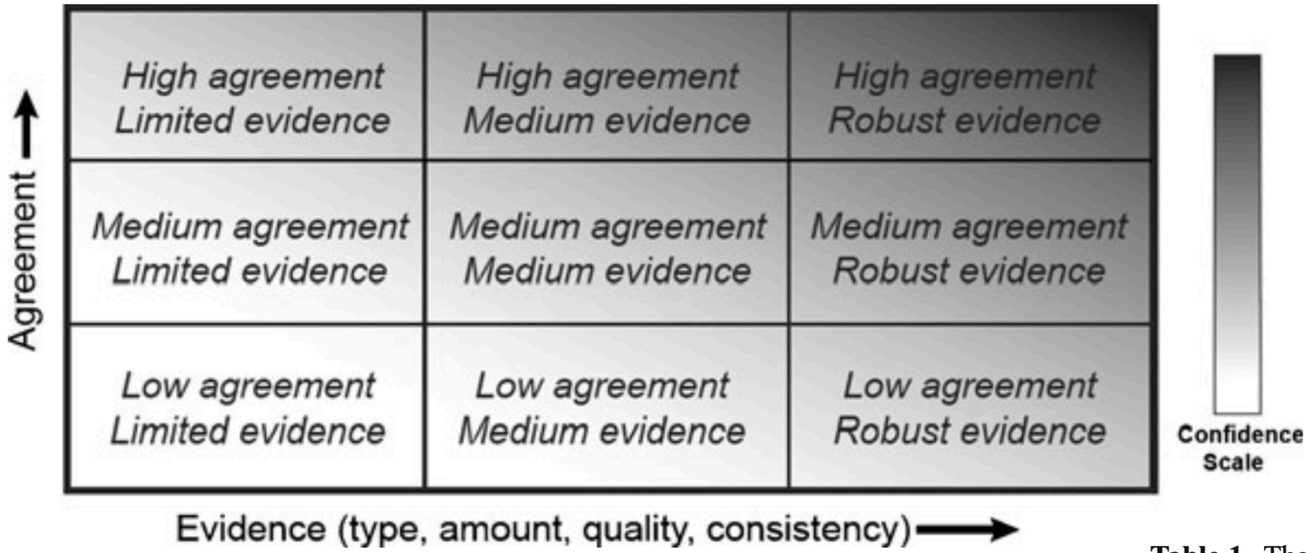
Empirical Software Engineering (2026) 31:4
https://doi.org/10.1007/s10664-025-10728-9

The unfinished business of evidence strength in software engineering: Current practices and future directions

Patrícia G. F. Matsubara[1] · Tayana Conte[2]

- Of 74 SE reviews citing Dybå & Dingsøyr:
  - only 12 tried a strength assessment, 22 claim one
  - GRADE in 10, all citing the 2004 version, not the more recent Core GRADE
  - 20 of 22 rated all findings together
  - CERQual used in only 1 review
  - only 2 of 22 matched method to review type
- Open question they state: Adopt GRADE/CERQual as is, adopt it, or create SE-specific approach?

And outside Medicine? IPCC (Climate Committee)



The IPCC AR5 guidance note on consistent treatment of uncertainties: a common approach across the working groups

Michael D. Mastrandrea · Katharine J. Mach · Gian-Kasper Plattner · Ottmar Edenhofer · Thomas F. Stocker · Christopher B. Field · Kristie L. Ebi · Patrick R. Matschoss

Table 1 The likelihood scale presented in the AR5 Guidance Note. Reproduced from Mastrandrea et al. (2010)

Term[a]	Likelihood of the outcome
Virtually certain	99–100% probability
Very likely	90–100% probability
Likely	66–100% probability
About as likely as not	33 to 66% probability
Unlikely	0–33% probability
Very unlikely	0–10% probability
Exceptionally unlikely	0–1% probability

And Social Sciences?

Use caution when applying behavioural science to policy

Authors: Hans IJzerman[1,2], Neil A. Lewis Jr.[3], Andrew K. Przybylski[4], Netta Weinstein[5], Lisa DeBruine[6], Stuart J. Ritchie[7], Simine Vazire[8,9], Patrick S. Forscher[1], Richard D. Morey[5], James D. Ivory[10], Farid Anvari[11]

"An ERL6 study built on a shaky ERL1 foundation will be of little use."

SE: a good systematic review only gets you to ERL3. Most SE advice to practitioners sits there or lower,

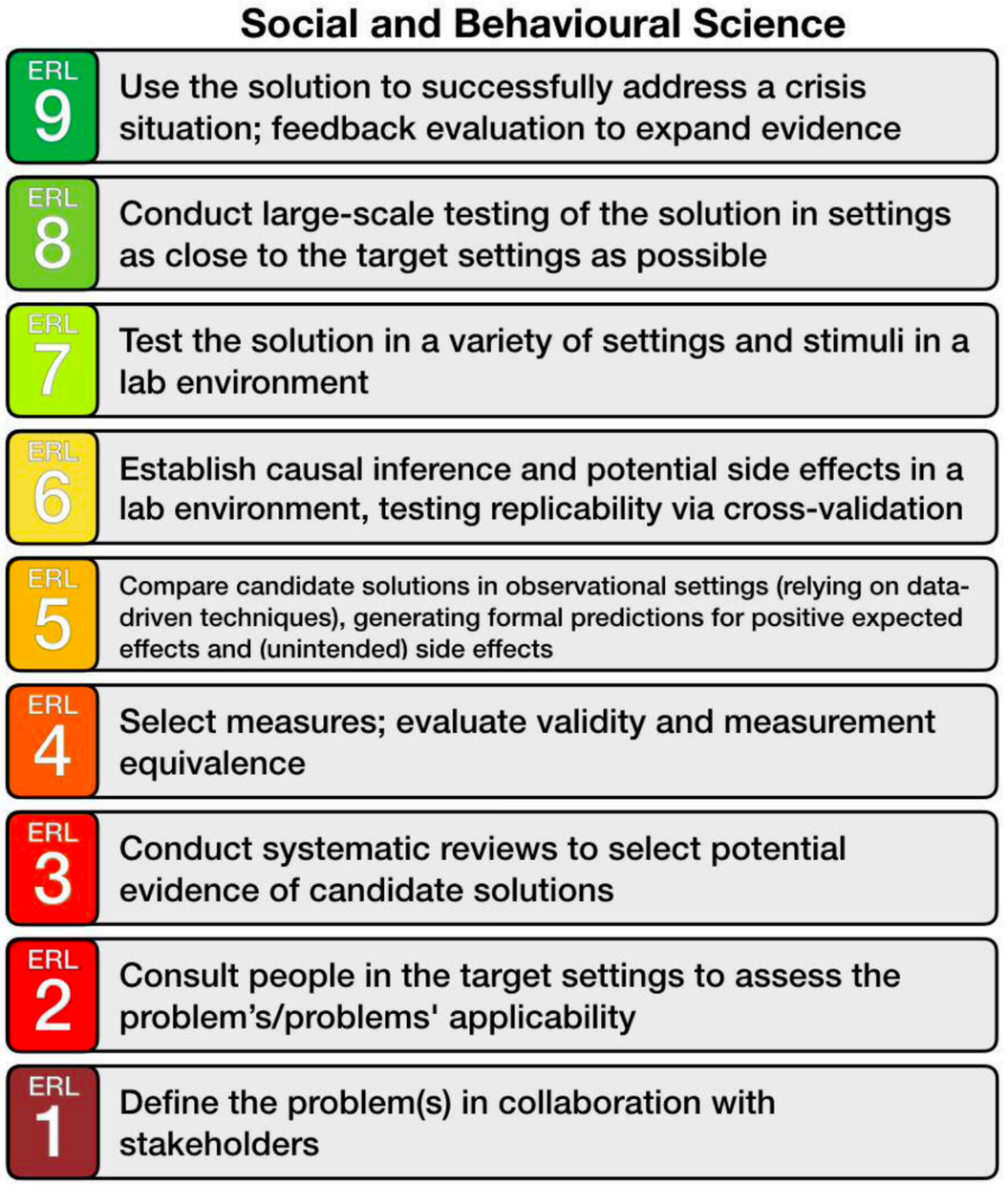


Figure 2. Proposed Social and Behavioural Sciences Evidence Readiness Levels

State of Evidence

- The current state of evidence in Software Engineering is still not strong
  - Practical relevance and impact?
  - Potential for transfer into practice and adoption?

- But there is hope…
  - Importance of empirical research recognised
    - Almost impossible to get into ICSE without lots of empirical data & analysis
  - Growth of a strong research community over last two decades
  - Lots of methodological papers and guidance

Summary: Evidence of Research (in SE)

- How strong is this piece of evidence, here?
  - Wohlin's evidence profile
  - Shaw’s validation types for papers (tomorrow)
  - Per-study quality checklists (Dybå&Dingsöyr 2008, Ivarsson&Gorschek rigour/relevance, Empirical Standards)
- How certain are we of a finding across studies?
  - GRADE (quantitative)
  - GRADE-CERQual (qualitative)
- Does the evidence actually support this claim, with the stated qualifier?
  - Toulmin argument model
  - Claim-evidence mapping (tomorrow)

NB! Quality of a study ≠ certainty of a finding

- Quality of a study ≠ certainty of a finding
  - Study quality is about one paper: was it designed, run and reported well?
  - Certainty of a finding about one finding across many studies: how sure are we it's true, here, for our purpose?
  - Quality is just one input to certainty. others span body of evidence (GRADE: inconsistency, indirectness, imprecision and publication bias)
- Why the two differ, some examples:
  - **Good studies, inconsistent results.** Five rigorous TDD experiments point in opposite directions, so "TDD improves quality" has low certainty.
  - **Good studies, indirect evidence.** Rigorous experiments with students on toy tasks are rated down for "professionals on real systems". This is Cartwright's "worked there vs will work here".
  - **Good studies, but too few or too selective.** Two excellent small studies give imprecise results, and publication bias links back to S2/S3.
  - **Imperfect studies, higher certainty.** Many independent, consistent, relevant sources can reach moderate or high confidence, as in CERQual's childbirth example (15 studies from 10 countries).

Science vs Engineering

	Objectives	Methods	Outcomes	Nature of Work
Science	Understand World Discover & Explain & Predict Phenomena	Scientific Method Rigorous Testing & Validation of Ideas	New Knowledge & Understanding Published: Theories + Models + Findings	Exploratory & Theoretical Hypothesis & Experimentation Answer: How? Why?
Engineering	Create, Design, Build & Improve Implement Practical Solutions to Real Problems	Design process Consider constraints (Cost, Safety, Regulations, Env)	Tangible Tech, Systems, Tools, & Processes Serve practical purposes & Improve life	Project-based & Application-focused Specific Criteria & Needs

Important vs Cited & How are you assessed?

Let's discuss!

- Think of two papers in your area that you consider important: one highly cited, one much less cited.

- Why might their citation counts differ?

- Does either represent better science?

- What counts when researchers are hired, promoted or funded where you work: number of papers, venues, citations, external funding, other things?

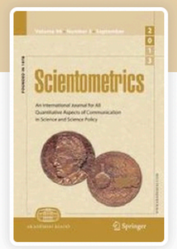
- Do you know whether your university or funder has signed DORA or CoARA, and does it make any difference in practice?

Definitions and background

- *Scientometrics* is the science of measuring and analyzing science.
- *Bibliometrics* is the science of measuring and analyzing publications.
- Scientometrics is often done by applying bibliometrics on scientific publications.

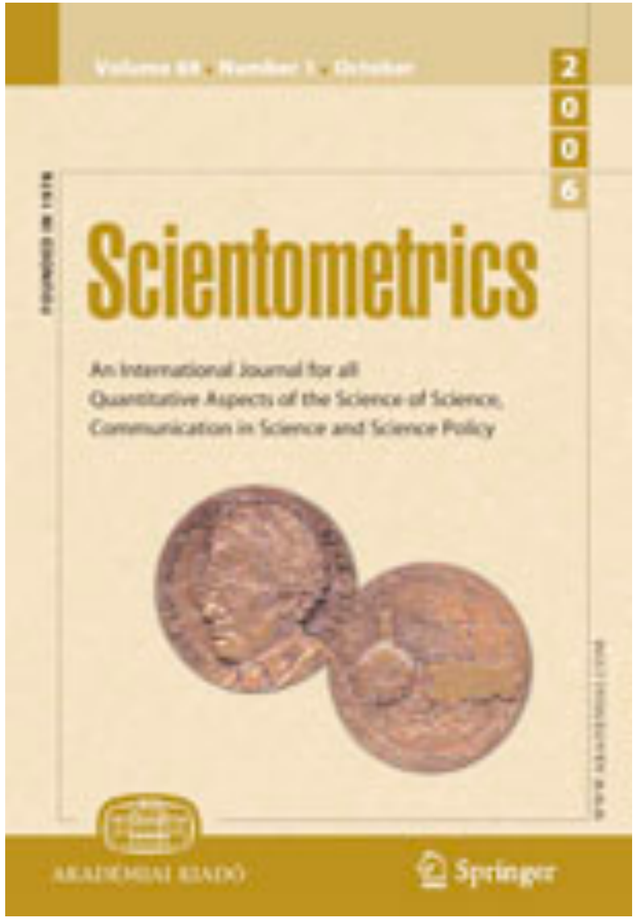
- There are of course scientific journals in this research area

Springer



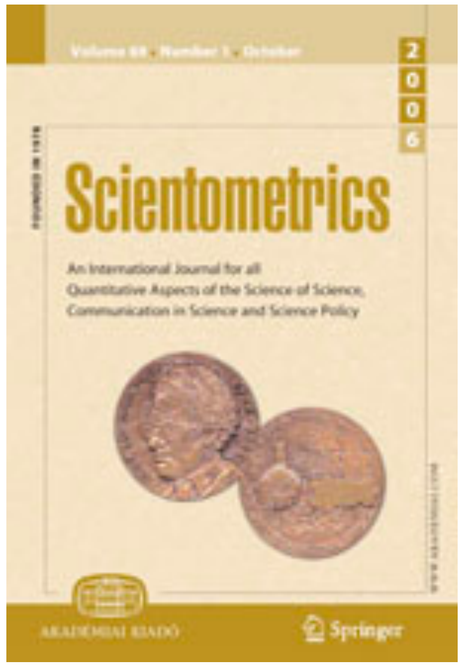
Scientometrics

An International Journal for all Quantitative Aspects of the Science of Science, Communication in Science and Science Policy



Example of papers in Scientometrics

- https://link.springer.com/journal/11192/volumes-and-issues/131-11
- e.g.: Claes Wohlin, “*A new index for the citation curve of researchers*”, Journal of Scientometrics, Vol 81, Num 2, November 2009. ☺



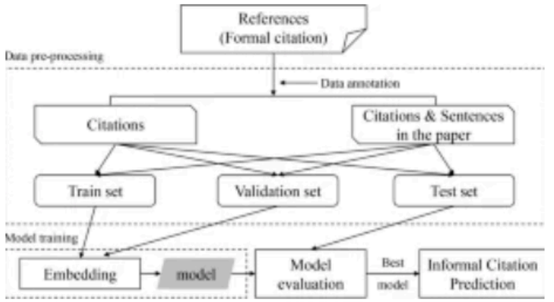
Papers in Scientometrics 2023

A study of BERT-based methods for formal citation identification of scientific data

Ning Yang, Zhiqiang Zhang & Feihu Huang

OriginalPaper | Published: 16 September 2023 |

Pages: 5865 - 5881



Do male and female authors employ different journal choice strategies?

Hayk Amirkhanyan, Michał Krawczyk & Maciej Wilamowski

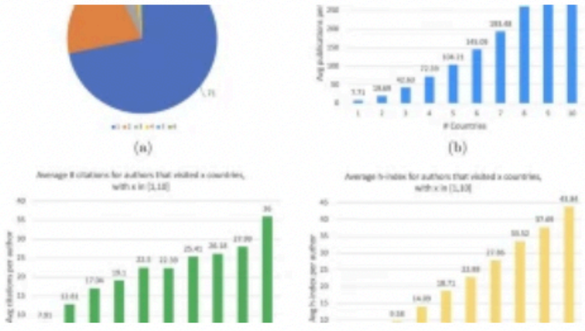
OriginalPaper | Open Access | Published: 10 October 2023 | Pages: 5905 - 5928

An analysis of international mobility and research productivity in computer science

Irene Finocchi, Andrea Ribichini & Marco Schaerf

OriginalPaper | Open Access |

Published: 25 September 2023 | Pages: 6147 - 6175



Motivation for metrics (University perspective)

- Several important reports during the last years:
  - ”Bibliometriska undersökningen”, Akademi Sydost
    - Used a number of different metrics and sources
  - ”Högskolerankingen”, Sydsvenska Industri- och Handelskammaren
    - One criterion (out of 8) was the number of publications (indexed by the ISI Web of Science)

Motivation for metrics (University perspective)

- ”Resurser för kvalitet”, Utbildningsdepartementet
  - Suggests that governmental research funds should be based on the quality of the research, measured by e.g. the number of citations
- Forskningsproppen 2012 and on
  - 10-20% of total research funding should be awarded based on “excellence”
  - Measured as both bibliometrics but also through expert assessment (international experts evaluate each major area)
  - Similar to UK’s REF which happens every 4-6 years

Motivation (University perspective)

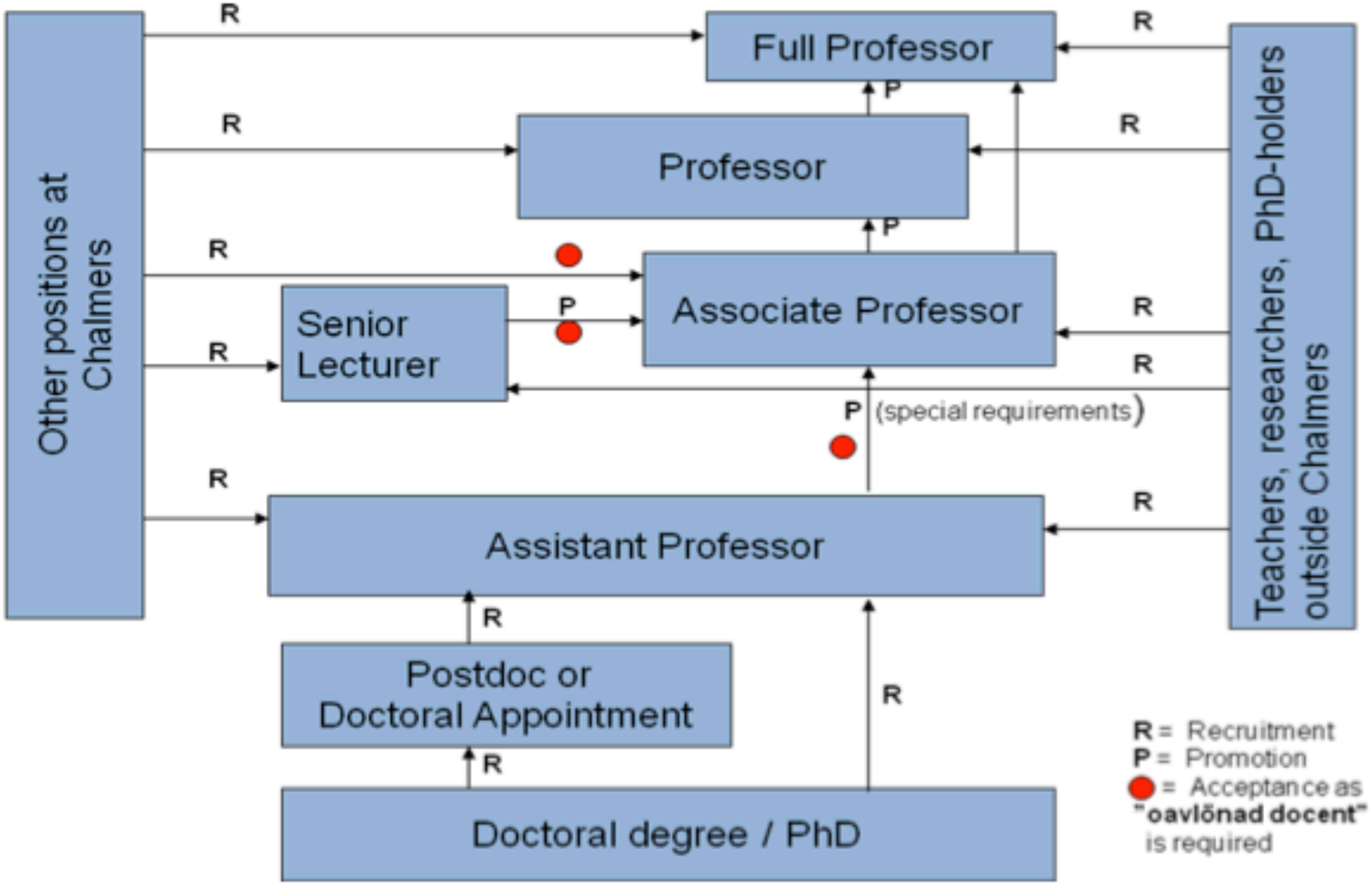
- Swedish Science Council (VR) wants everyone to state number of citations for every listed paper, but also says:

”Bibliometriska data, till exempel i form av det totala antalet citeringar för en sökande, bör användas med försiktighet i beredningsarbetet. Det har visat sig att sådana kan bli missvisande, om inte ett omfattande arbete med kvalitetssäkring utförts. Dock ska de sökande fr o m 2009 själva rapportera antalet citeringar för sina publikationer i ansökan.”

Motivation (individual perspective)

- Better understanding of ”system”
  - How are researchers being evaluated/ranked?
  - How are journals and conferences ranked?
- Can be useful e.g. when
  - selecting where to publish your research results
  - improving the impact of your research
  - applying for positions
  - applying for project funding
- However, trend seems to be somewhat away from explicit use of metrics
  - But implicit use still abundant or even seen as the normal state!?

Motivation (individual perspective)



University appointment regulations for teaching and research faculty at Chalmers

2011

Motivation (individual perspective)

4.2 Fundamental qualification requirements

Scientific and pedagogical expertise is required for each academic position at Chalmers. Other experience and abilities are also required, depending on the needs of each department with regard to the subject content and responsibilities of the position. For some positions, the requirement for scientific expertise may be replaced by other specific alternative professional skills, for example artistic, pedagogical or leadership abilities, see Section 4.2.1 and Sections 6.1 and 6.2.

In addition to fulfilling the formal scientific and pedagogical qualifications, promotion requires compliance with *Chalmers' Strategies* with regard to responsibility and involvement, respect for others and equality of treatment.

University appointment regulations for teaching and research faculty at Chalmers

2011

Motivation (individual perspective)

4.2.1 Scientific expertise

Scientific expertise shall be demonstrated through one's own research and the planning and leadership of research. It can also be shown by the ability to achieve results through cooperation with other researchers inside or outside Chalmers. The leading international level of research in a specialist field shall be the reference point for assessment of scientific expertise.

In architecture and design, artistic expertise can be equivalent to scientific expertise.

University appointment regulations for teaching and research faculty at Chalmers 2011

Motivation (individual perspective)

4.2.2 Pedagogical expertise

Pedagogical expertise should be shown through one's own teaching and the ability to generate commitment and interest in the subject, to organise knowledge in well-structured and highly esteemed courses, to motivate students in their learning, and to communicate with students and other teachers. The abilities to hold a comprehensive view and to engage in renewal are valuable assets. Pedagogical expertise is founded on sound and extensive knowledge of the subject in question and a reflective attitude both towards one's own pedagogical work and student learning. Links with research in the subject are also important in relation to pedagogical expertise. Pedagogical qualifications should be documented in a Pedagogical Portfolio.

University appointment regulations for teaching and research faculty at Chalmers 2011

Docent at Chalmers: What is assessed?

- **1. Scientific independence**: a documented ability to formulate and pursue your own research questions. You need **substantial personal contributions** and an **independent line of research beyond your PhD** and your former supervisors' work.
- **2. Scientific contributions**: research of sufficient **originality, significance, quality, breadth, depth and volume beyond the doctoral level**, judged in light of the field's conditions.
- **3. Teaching skill**: the ability to teach and supervise, especially at the advanced and doctoral levels, plus relevant training in university teaching.
- *Publication counts alone are not enough.*
  - The formal rules (handläggningsordningen) set only a minimum.
- Can strengthen the case: leadership, collaboration with society and industry, and other academic merits.
- Main evidence for **independence**: documented scientific work.

Motivation (individual perspective)

- Even internal money at Chalmers will be distributed partly according to success in publishing in the “right” fora
  - “per head” models are disappearing
  - “per (strong) paper” models are coming

Metrics (scientists)

- Total number of publications
- Total number of citations
- Average number of citations per paper
- H-index (proposed by Hirsch in 2005)
  - An h-index of X means that a person have X publications that have X citations or more
  - This metric is useful because it discounts the disproportionate weight of highly cited papers or papers that have not yet been cited.
  - Combines *quantity* (number of publications) and *quality* (impact/citations to these publications)
- The mean number of citations to offline articles is 2.74, whereas the mean number of citations to online articles is 7.03!

  (Lawrence, 2001)

Metric for publications: Impact Factor et al

- The average number of cites to articles in a journal/proceedings in the last 2 years (-1)
- PageRank (similar to Google's algorithm) is a recursive impact factor, to give citations from journals that have high impact greater weight than citations from low-impact journals
- The number of citations to papers in a particular journal does not really directly measure the true quality of a journal, much less the scientific merit of the papers within it.
- Cannot be compared directly between different research areas/disciplines. It depends on, e.g.:
  - the absolute number of researchers in the area,
  - the average number of authors on each paper,
  - the nature of results in different research areas,
  - citation habits between different disciplines, particularly the number of citations per paper
- Review/survey papers/journals always get a lot of citations
- Venue h-index (like GScholar's area pages) depend heavily on size of venue
- Luckily, trend seems to be away from venue-specific rankings
  - But many countries still have requirements on publication in specific "quality tiers"

Sources for SE

- Journal of Systems and Software (JSS) yearly ranking
  - Glass et al since 1996, 2011 one covers 2004-2008
  - 1.0 points if 1 au, 0.7/au for 2 au, 0.5/au for 3, 0.4/au for >3
  - Originally based on only 7 journals (TSE, TOSEM, JSS, SPE, EMSE, IST, IEEE SW)
    - But has evolved over the years, also including conferences nowadays
    - Infrequent updates
- Feldt had/has summary lists:
  - http://www.cse.chalmers.se/~feldt/advice/isi_listed_se_journals.html
  - Thomson Reuters threatened legal action
  - Alternative one not updated since 2016

ISI JOURNALS - SOFTWARE ENGINEERING

Journal	11	10	09	08	JSS
1. IEEE Transactions on Software Engineering (TSE)	**1.98**	2.22	3.75	3.57	Yes
2. Communications of the ACM (CACM)	**1.92**	2.35	2.35	2.65	
3. Empirical Software Engineering (ESEJ)	**1.85**	1.78	1.61	1.09	Yes
4. IEEE Software (SW)	**1.51**	1.51	2.04	2.10	Yes
5. IEEE Computer (Comp)	**1.47**	1.79	2.21	2.09	
6. ACM Transactions on Software Engineering and Methodology (TOSEM)	**1.27**	1.69	2.03	3.96	Yes
7. Information and Software Technology (IST)	**1.25**	1.51	1.82	1.20	Yes
8. Software and Systems Modeling (SoSyM)	**1.06**	1.27	1.53	N/A	
9. Requirements Engineering Journal (REJ)	**0.97**	0.86	0.93	1.63	
10. Software Testing Verification & Reliability (STVR)	**0.96**	0.76	1.63	1.05	
11. Automated Software Engineering (ASE)	**0.86**	0.81	1.27	N/A	
12. Journal of Systems and Software (JSS)	**0.84**	1.28	1.34	1.24	Yes
13. Software Maintenance and Evolution - Research & Practice (SMERP)	**0.84**	0.61	1.14	0.97	
14. IBM Journal of Research and Development (IBM JRD, IBM Systems Journal was merged into this one in 2009)	**0.72**	1.79	1.29	1.88	
15. Software Practice & Experience (SPE)	**0.52**	0.57	0.67	0.71	Yes
16. Software Quality Journal (SQJ)	**0.42**	0.75	0.98	0.95	
17. IET Software (IET SW, was called 'IEE Proceedings - Software' before 2007)	**0.33**	0.67	0.65	0.54	

- Core general recommendation: "**do not use journal-based metrics such as the JIF as a surrogate** for the quality of individual articles, individual scientists' contributions, or hiring, promotion and funding decisions”
- 17 detailed recommendations addressed to 5 audiences, examples:
  - Funders and institutions:
    - be explicit about criteria; **content > metrics or venue**, especially for early-career researchers
    - value **all outputs, including datasets and software**
  - Publishers:
    - offer and promote article-level metrics
    - require author-contribution statements (CRediT)
  - Researchers:
    - judge by content when on committees;
    - use a range of article-level indicators
- Limits of DORA:
  - a declaration, with no enforcement or monitoring (CoARA tries to address this)

CoARA
Coalition for Advancing Research Assessment

- CoARA = Coalition for Advancing Research Assessment
- https://www.coara.org/
- Assessment should recognise **diverse** outputs, practices and activities.
- It should be based **primarily on qualitative judgement, with peer review central**, supported by responsible use of indicators.
- Timeline: an action plan within **1 year** of signing; full review cycle within **5 years.**

Universitetet i Bergen
Website

Oslo Metropolitan University
Website

The University of Oslo
Website

Karolinska Institutet
Website

Mid Sweden University
Website

Halmstad University
Website

Norwegian University of Science and Technology (NTNU)
Website

University of South-Eastern Norway
Website

University of Agder
Website

University of Gothenburg
Website

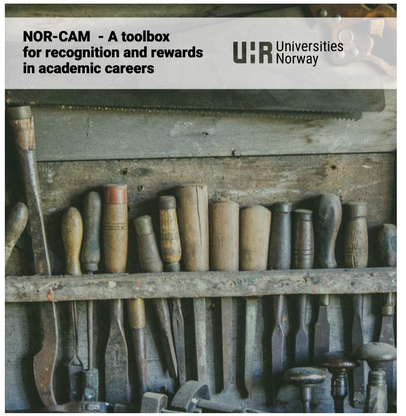
Jönköping University
Website

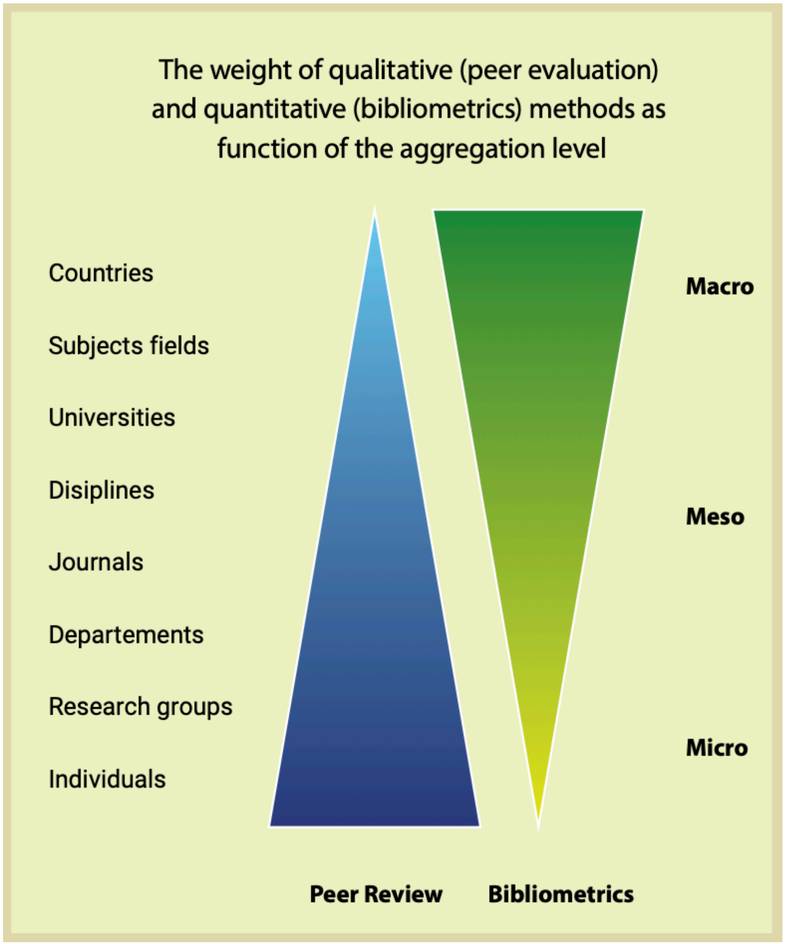
Luleå University of Technology
Website

NOR-CAM - A toolbox for recognition and rewards in academic careers

Universities Norway

- Bibliometrics must not be used in isolation
- Bibliometrics do not look ahead
- Bibliometrics do not understand varying contexts
- Bibliometrics cannot make decisions





Kanalregisteret

Academic publishing is no longer part of the funding of higher education institutions in Norway

From 2025 the Norwegian Government makes changes in the system for funding of Universities and University Colleges. Publication points will no longer be part of the funding of the higher education institutions in Norway.

Published: 18. September 2024

Meanwhile in Sweden… ;)

Vetenskapsrådet

På svenska

Start / Just now / News

NEWS

Publications and external funding to play a greater role in direct government grants

The Government is changing its model for allocating direct grants to higher education institutions (HEIs). The impact of the research and the HEIs' ability to attract external funding will be rewarded more than previously. The Swedish Research Council has been tasked by the Government to develop a proposal ahead of the change, which will come into force in 2026.

Teaser for tomorrow: Good Papers, Publishing, Reviewing, a Peer Review Crisis

Let's discuss!

- Have you had a paper desk rejected or rejected after review?

- If so, what did you learn from it?

- If not, what would you want to receive from the venue if it happened?

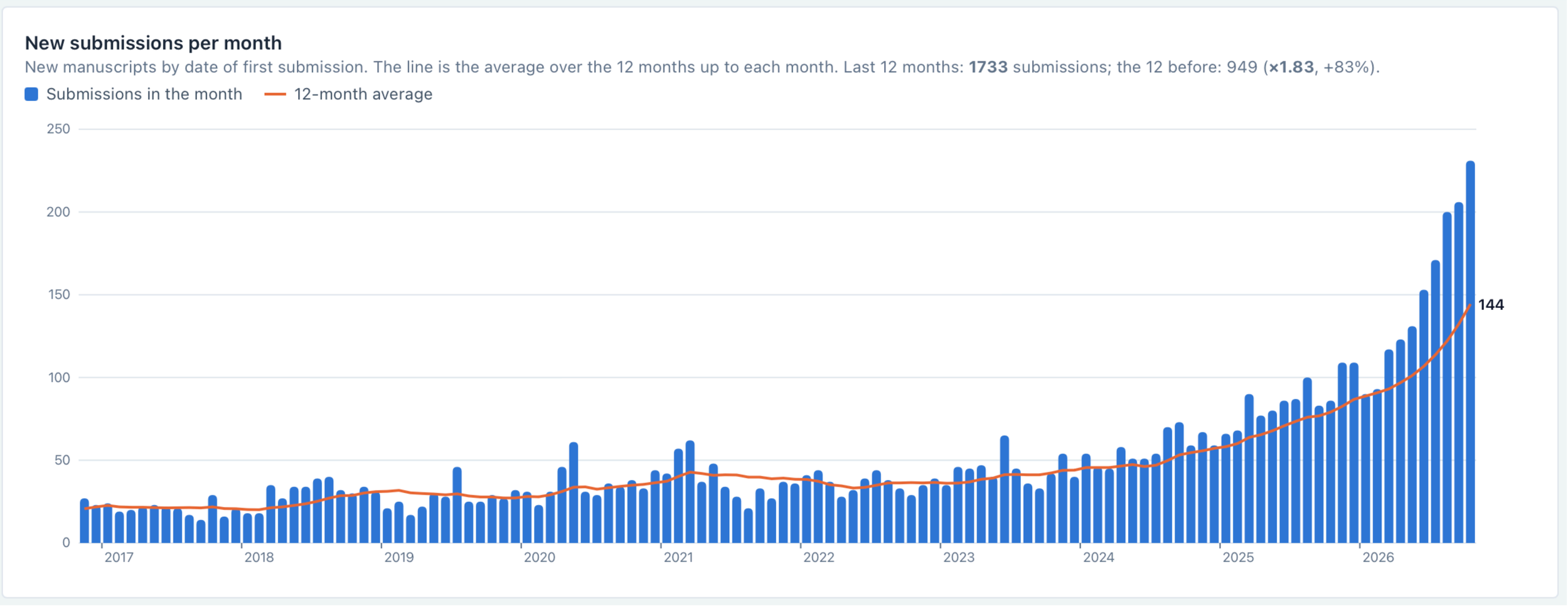
A record year

ESEM 2026 Technical Track

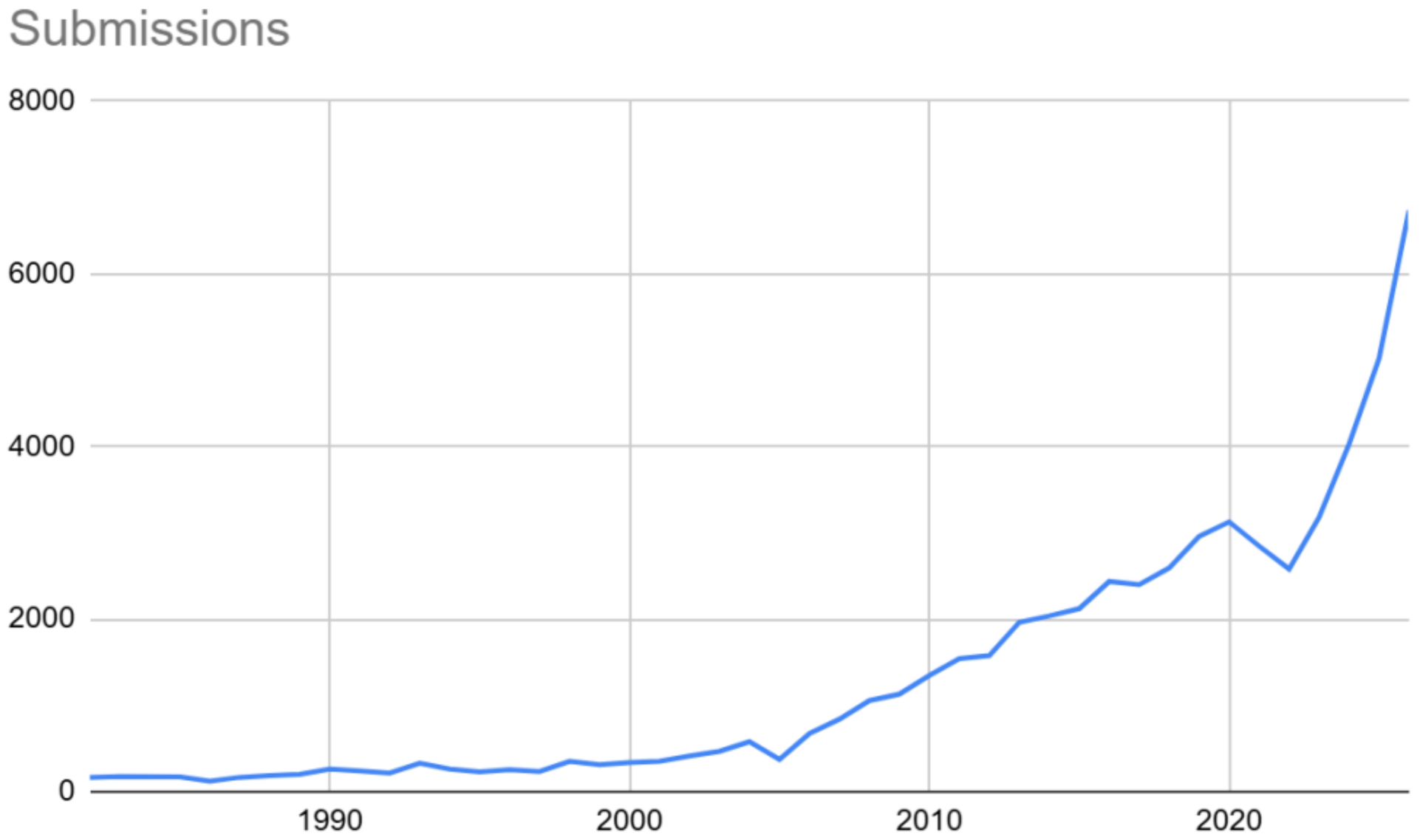
	2024	2025	2026	2026 vs 2025	ICSE research track 2026 vs 2025
Submitted full papers	149	154	**268**	**+74%**	**+28%** 1,150 to 1,469
Reviewed (after desk rejects)	139	145	**240**	**+66%**	n/a
Accepted	34	28	**51**	**+82%**	**+31%** 245 to 321
Accept rate (of reviewed)	24%	19%	**21%**	**+1.9 pts**	**21.3% to 21.9%** +0.6 pts

If I was PC chair next year I would count on at least +50% also next year!

EMSE Submissions per Month



CHI Submissions



AI Submissions